

Red Tide — Visual Overhaul Spec

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Purpose

This spec rebuilds every visual in Red Tide with Blender: the four divers, every weapon, all ~200 beasts and flora, the five enemy factions and five bosses, and all five maps.

Systems, numbers, and rules in the Red Tide design doc (Arcade Game 5) stay exactly as they are; this changes how the game looks, never how it plays.

What it supersedes in the design doc.

- The "Presentation and technology" opening and the "Art" section: flat-shaded procedural meshes, CreatureBuilder lathe bodies, and "every mesh, texture, and animation generated in code".
- Depth's no-assets rule as it applies to Red Tide visuals. Section 3 sets the new rule: models, textures, and animations are authored in Blender by Claude Code through the connected Blender tool, saved as committed source files and Python build scripts, and exported to glTF for raylib.
- "Weapons and hands" (procedural meshes) and "Maps are built in code from layout data" as the source of geometry. Layout data still defines where rooms, racks, and habitats go; Blender builds what they look like.

What stays. Raylib as the engine, the ecosystem and its species records, the layout data, the ink identity of Depth (refined, see section 2), the brass-helmet HUD, the dossiers, and every tell and counter. A beast's species record keeps driving its colours and parameters; Blender now turns those into a properly modelled creature.

Relationship to The Trawl. The Trawl Visual Overhaul Spec covers the same ground for the trawler game. Red Tide and The Trawl share the engine, so build these once in shared code and use them in both: the PBR and lighting renderer, the humanoid rig, the first-person arms and viewmodel system, the gun-part kit and materials, the fish rigs, and the screenshot harness. Where the two specs overlap, follow whichever was built first and keep them consistent.

How Claude Code should use it. Work in the phases in section 10. Reread the relevant section before each phase. After each phase, render the screenshot set and check it against the acceptance criteria. If something here is impossible in raylib or too expensive,

say so with numbers and propose an alternative instead of quietly doing less.

Art direction

The target is "a living underwater world you could screenshot as a postcard, then get eaten in": rich stylized-realistic creatures and places, heavy period diving gear, and guns that look machined and real, all under water that has depth, colour, and light.

Tone by map. Red Tide runs brighter than The Trawl, but each map has its own mood, and the mood darkens as the maps get deeper.

Map	Mood	Light	Signature colours
The Sunken Ship	Eerie, claustrophobic, rich decay	Green-blue shafts through portholes; work lamps	Rusted iron, rotted mahogany, tarnished brass, green-blue water
The Underwater Cave	Dark, echoing, mineral	Single light shafts at the Mouth; diver lamps; bioluminescence; air-pocket shimmer	Wet black stone, cold teal, crystal coral violet and white
The Coral Reef	The bright map: vivid, sunlit, deceptively friendly	Strong sun, caustics everywhere, deep blue beyond the drop-off	Saturated coral pinks, oranges, and purples, turquoise water, white sand
Atlantis	Grand, mournful, ancient	Filtered blue from above, glow-moss and crystal-engine light at the summit	Pale marble with green algae, bronze and gold, deep blue streets
Approaching the Void	Cosmic dread, the deepest and darkest	Almost none: diver lamps, station lights, bioluminescence, the pale light far below	Black sand, rust-orange station, cold white and violet bioluminescence

Era and material language. Divers and factions wear 1890s–1920s diving technology: copper and brass helmets, canvas and rubber suits, riveted hard suits, hoses, lead boots. Guns are period-built but underwater: gas reservoirs, glass needle cartridges, sealed

brass clips, steel spears. Atlantis is ancient (bronze, marble, mosaic); the Void station is early-20th-century industrial with something alien seeping in. Nothing modern: no polymer, no LEDs, no modern scuba fins.

Three levels of realism.

1. **Realistic layer: guns and diving gear.** Accurate mechanical parts, true materials (brass, copper, blued steel, glass, rubber, leather, walnut), wear and corrosion from salt water. The gun in your hands is the most detailed object on screen.
2. **Stylized-realistic layer: creatures, divers' bodies, enemies, bosses.** Anatomically believable, real species read at a glance, slightly exaggerated features for readability (bigger eyes on prey, bigger teeth on predators), detailed skin, scales, and shells.
3. **Stylized layer: environments.** Rich and detailed but painterly: clean big shapes, then dense small detail (coral, weed, barnacles, debris) that sells scale.

The Depth ink identity, refined (decided). Keep the ink line: a thin depth-and-normal edge line, coloured by the water tint rather than black, fading with distance and fog, so a screenshot still reads as Depth. The Blender-built detail takes priority over the line: wherever the line hides or flattens modelled detail (bevels, faces, gun parts, scale patterns), thin it or drop it locally. Tune line weight per asset class so guns and faces stay clean. The Bayer stipple is off by default and kept only as a settings toggle, since it fights the baked texture detail. Both get settings toggles.

Readability rules.

- Every creature's size class, threat, and weak point read at a glance (the design doc's weak points: gills, eyes, underside, vent, throat get subtle visual emphasis: lighter, softer, or veined).
- Divers always out-silhouette everything: four fixed shapes, visible at 40 m in clear water.
- Enemies never look like beasts, and beasts never look like decor. Camouflaged ambushers are hidden on purpose and get a subtle shimmer tell up close.
- Blood is the game's central resource, so it must always read: dark red ribbons in clear water, near-black in the deep, never orange or pink.

Blender pipeline

Claude Code builds every asset in Blender through the connected Blender tool, but every asset must also be reproducible from a committed Python script, so nothing lives only inside a .blend file that no one can regenerate.

Working method.

1. Use the live Blender connection to model, inspect, and iterate: build the object, render a preview from several angles, look at the render, fix, repeat.
2. When an asset is approved, capture the steps as a Blender Python script (`tools/blender/<category>/<asset>.py`) that rebuilds it from scratch, and commit both the script and the resulting .blend file.
3. Export to glTF 2.0 binary (.glb) with skins and animations into `assets/redtide/` (map and creature content) or `assets/shared/` (renderer test assets, diver rig, arms, gun kit, shared fish).
4. A single command (`tools/blender/build_all.py` , run with Blender headless) rebuilds every asset, so a change to a shared part (for example, the brass helmet) can be pushed to every diver and skin.

Shared building blocks. Write these once and reuse them everywhere:

- **Material library** (one .blend appended by all others): brass, copper with verdigris, blued and bright steel, rusted iron, walnut, rubber, canvas, leather, glass, marble, sandstone, coral, sand, scale and skin types. Every material bakes to a PBR set.
- **Creature kit**: one rig per body plan from the design doc (fusiform fish, flat fish, eel, ray, shark, turtle, crocodilian, crab, shrimp, cephalopod, jelly, worm, amphibian, seal, whale), plus a script that reads a species record (size, colours, fins, spines, eyes, tags) and adjusts the base mesh for that species. Unique beasts and bosses are modelled individually on top of the kit.
- **Humanoid kit**: one diver rig shared with The Trawl, with outfit layers per diver and per faction.
- **Gun-part kit**: reusable barrels, cylinders, triggers, screws, sights, gas reservoirs, and stocks, so new guns are assembled from proven parts.
- **Modular map kits** per map (hull plates and decks, cave rock and formations, coral pieces, Atlantis building parts, station modules), placed by the existing layout data.

Technique standards.

- Bevel every hard edge (bevel modifier, 0.5–2 mm on guns, 1–5 cm on architecture) and use weighted normals. Edge highlights are most of what makes models look finished.
- Model high-poly detail, then bake normal, AO, and curvature maps onto the game mesh.

- Texture with Blender shader nodes (noise, Voronoi, musgrave, curvature-driven wear and grime), then bake to images. No downloaded textures, purchased assets, or AI-generated images.
- Clean UVs with consistent texel density, packed with trim sheets for architecture.
- Rig with Rigify-style skeletons; animate with keyframes plus procedural shape keys. Bake every action to the glTF before export.
- Three LODs per creature, character, and gun (the design doc's "3 per creature; billboards past 80 m" stands), generated with decimate and checked by eye.

Raylib integration. Raylib loads glTF meshes, materials, and skeletal animations (`LoadModel` , `LoadModelAnimations`). Write custom PBR and water shaders, add GPU skinning and instancing (one draw per species per LOD, as the budget requires), and convert .glb to a faster engine format at build time if load times demand it.

Budgets (LOD0).

Asset	Triangles	Textures
First-person gun	10,000–25,000	2048 ² PBR set
First-person arms (per diver)	6,000–10,000	1024 ²
Diver, full body	12,000–20,000	2048 ² suit + 1024 ² helmet
Faction enemy	8,000–15,000	1024 ²
Boss (size 6)	40,000–80,000	4096 ² or several 2048 ²
Beast, size 4–5	6,000–15,000	1024 ²
Beast, size 1–3	500–4,000	256–512 ²
Schooling fish	150–500	Shared atlas
Flora and coral pieces	100–3,000	Shared atlases
Map modules	Per the per-map draw-call budget	Trim sheets

Player characters: the four divers

Each diver is a fully modelled person inside heavy, believable period diving gear, with a silhouette that no skin can change, as the design doc requires. Gear is the realistic layer, so helmets, suits, and fittings get gun-level care.

The four suits

Diver	Silhouette (never changes)	Gear to model	Personality in posture (from the quips)
The Diver	Round brass helmet, classic standard dress	Copper-and-brass helmet with three ports and grilles, a bolted corselet, canvas-and-rubber dress, weighted belt, lead-soled boots, an air hose looping from the back, a sheath knife	Dry: economical, upright, unhurried
The Whaler	Hooded oilskin slicker over a lighter rig	Hood drawn over a smaller face-plate helmet, long slicker with straps and toggles, harpoon-line coil, rope belt, tarred gloves	Gruff: heavy-shouldered, forward-leaning
The Stowaway	Patched, mismatched, bottle on a cord	A suit patched in different canvases and rubber, a dented helmet with one mismatched port, a bottle swinging on a cord at the chest, a sack, string-tied boots	Cheerful and slightly drunk: loose, bouncy, a little sway

Diver	Silhouette (never changes)	Gear to model	Personality in posture (from the quips)
The Mechanic	Riveted hard suit, boxy and armoured	Riveted steel atmospheric suit with articulated joints, a domed helmet with a grid of small ports, gauges and valves on the chest, a tool roll, pincer-glove option	Nervous and precise: tight, tidy, quick small movements

Bodies and faces

- One shared humanoid rig and body (about 7 heads tall), shared with The Trawl. The body exists under every suit so suits fold and move like clothing over a person.
- Faces are visible through helmet ports, so model them: brow, eyes with wet highlights and blinks, nose, mouth, stubble or beards. Expressions as shape keys (neutral, strain, fear, grin, shout). The mouth moves with voice chat and quips.
- Fogged or wet glass and reflections on the helmet port glass, so the face reads through it without looking like an empty hole.

Suit materials and wear

- Brass and copper with verdigris collecting in seams and around rivets, polished bright where hands touch.
- Canvas that darkens and stiffens with water, rubber with cracks at folds, leather straps with stitching, steel with rust bloom at joints.
- Grime and blood build up through a match: blood clouds briefly from wounds and fouls the suit, then thins.

Skins

The design doc's token skins (verdigris, red tide, bone, pearl, Atlantean) change materials and colours only, never the silhouette. Build each as a material set on the same mesh: verdigris (green patina, aged), red tide (deep red dyed canvas, rust-red brass), bone (bleached ivory plates, scrimshaw), pearl (nacre plating, iridescent), and Atlantean (bronze with gold inlay and Atlantis motifs).

Animation

- **Swimming.** Underwater locomotion, not land walking: slow heavy strokes, legs kicking, the body pitching with direction. Heavy-boot walking on the bottom where the design doc has floors. Drifting out of a slipstream.
- **Air pockets.** Head and shoulders above the surface, wet suit and dripping in the air, a different breathing rhythm.
- **Combat.** Each weapon class's hold, aim, fire, and reload in third person (matching first person), melee swings that fight water resistance (slow windup, fast strike).
- **Downs and revives.** Downed divers drift and sink slowly, a hand pressed to the wound, firing the Last Stand pistol. Revives show a teammate grabbing the harness and working a valve.
- **States.** Netted, grabbed by tentacles, swallowed (the Goliath's Inhale), parasite attached, stunned, poisoned (a green tint to the breath bubbles), and death.
- **Quips and pings.** Head turns to the teammate who is talking; a gesture accompanies callbacks.
- Bubbles leave every helmet exhaust on each breath. Breathing speeds up when the predator pulse rises.

First person

- The view is from inside the helmet: the brass HUD from the design doc frames the screen, with faint glass reflections, droplets after an air pocket, and condensation when breathing fast.
- First-person arms wear that diver's gloves and sleeves (rubber gauntlets, tarred gloves, mismatched mitts, or the Mechanic's armoured sleeves).
- Every held item is gripped correctly with both hands where it should be: fingers wrapped around grips and fore-ends, thumbs on hammers, a hand on the cocking lever of spearguns.
- Looking down shows the diver's own chest gear and boots.
- Tonic drinking, charm use, buying, building, and reviving all have first-person hand animations. Drinking a tonic underwater: the bottle feeds through a valve in the helmet.

Weapons

Red Tide’s guns are invented underwater weapons, so their realism comes from engineering logic: every gun must look like a machine someone could have built in 1910 to fire under 30 m of water, with every part that would need to exist (gas reservoirs, seals, valves, cartridges, cocking mechanisms) modelled and animated.

Construction standard

- Model each gun as an assembly of real parts: barrel with a bore and muzzle, receiver, trigger and guard, sights, grips and stocks, screws and pins, and the underwater systems: gas bottles with valves and gauges, rubber seals and O-rings at every joint, drain ports, sealed breech covers, and lanyard rings.
- Moving parts are separate, pivoted objects: cylinders turn, hammers cock, levers and bolts work, spears slide out, gauges’ needles drop as gas is used, glass cartridges show their remaining needles.
- True scale in centimetres, then framed by the viewmodel FOV.
- Bevels on every edge (0.5–2 mm); weighted normals; baked normal, AO, and curvature maps from a high-poly version.

Materials

Material	Look	Where
Brass and bronze	Warm gold, darkened in recesses, verdigris in seams	Gas reservoirs, fittings, sealed clips, the Gannet family
Blued and parkerized steel	Blue-black or grey, worn bright at edges	Barrels, receivers, the powder guns
Glass	Thick, slightly green, with bubbles inside and refraction	Needler cartridges, gauges, the Galvanic Rod’s cells
Rubber and gutta-percha	Matte black-brown, cracked at folds	Seals, grips, bladders, hoses
Oiled walnut, teak	Grain and figure, darker and swollen where wet	Stocks and fore-ends
Salt corrosion	Pitting, white salt bloom, barnacle spots at the muzzle	Generated from curvature and AO on all metal

Wear, salt bloom, and handling polish are procedural (curvature, AO, and grip masks). Every gun carries invented maker's stamps, serial numbers, and pressure-test marks. Never real trademarks.

Per-weapon design

Weapon	Visual concept	Key parts to model
The Cormorant (start pistol)	A compact brass gas revolver	Six-dart cylinder that swings out, a small gas bulb in the grip, a pressure needle on the side; darts with fins visible in the cylinder
Gannet	Bigger brother of the Cormorant	Ten-dart rotary magazine, a longer barrel with a bubble-vent shroud, a screw-in gas cartridge under the barrel
Twin Gannets	Two Gannets with mirrored grips	Both hands visible, alternating fire and reload
Needler Mk I and Mk II	A needle submachine gun fed from a glass cartridge	Glass cartridge of steel needles on top or side, a gas bottle stock, a cooling-and-bubble shroud; the Mk II longer with a drum-shaped cartridge
Sea-Pattern Carbine	A period service carbine sealed for water	Rubber-sealed breech, brass sealed clips that eject and sink, a barrel band, a rubber muzzle cap on a lanyard
Trawlerman's Rifle	A heavier sealed powder rifle	Longer barrel, bolt with a rubber-booted handle, a box magazine of sealed brass clips, ladder sight
Stormlock	A powder automatic rifle	Finned barrel, side-mounted magazine of sealed clips, a pistol grip and bipod-style fins

Weapon	Visual concept	Key parts to model
Flechette 12 and Drum Flechette	A pump scatter gun loaded with flechette shells	Tube magazine and loading gate, shells shown going in one by one; the Drum version with a round drum and a heavier frame
Long Speargun	A long rubber-band or pneumatic speargun	Wooden or steel stock, a steel spear with barbs and a line, a cocking strain visible on both arms
Bolt Harpoon	Heavy harpoon rifle	A side quiver of bolts, a winch crank, a scope with a brass tube; bolts pinned in walls stay visible until retrieved
Reef Rattler	A three-round burst needle rifle	Three-chambered glass cartridges, a striker that clicks visibly three times
Gatling Needler	A crank-and-motor rotary needle gun	Rotating barrel cluster that spins up for 0.5 s, a large glass hopper, a gas tank on a back strap
Cannon Harpoon	A shoulder harpoon launcher	Explosive harpoon head with a fuse, a recoil spring, a heavy shoulder pad
Limpet Launcher	A drum launcher for sticky charges	Four-round drum, charges with magnetic or barnacle-glue pads and a blinking fuse light
Net Gun	A wide-mouth launcher	A folded net visible in the muzzle with weights at its corners
Chumthrower	A pump sprayer for chum	A glass and brass tank showing the chum sloshing inside, a hand pump, a nozzle

Weapon	Visual concept	Key parts to model
Harpoon Cannon (Harpoon Hour)	A shoulder-mounted salvage cannon	Massive barrel, a feed of bolts, an exaggerated recoil that shoves the diver back

Melee weapons get the same care: the diver's knife (a stainless blade with a rubber grip and a sheath), the Boarding Axe (a rusted steel head on a tarred haft), the Tesla Gaff (a hook with copper coils and a battery), the Trident (a bronze trident with barbs), and the Sawtooth (a real sawfish rostrum with teeth, bound to a handle).

Wonder weapons

Each map's wonder weapon is the most spectacular object on that map, and its look should explain what it does.

- **The Galvanic Rod (Ship).** A ship's dynamo on a pole: brass housing, copper windings, a hand crank, arcs crawling over the coils. Forged as the Lightning Keel with a ship's keel plate and a constant crackle.
- **The Resonator (Cave).** A sonic cannon of tuned crystal coral and brass horns; it visibly vibrates and rings. Forged as the Deep Bell, with a bronze bell at the muzzle.
- **The Anemone Gun (Reef).** A living weapon: coral and shell grown around a brass core, with polyps pulsing in the barrel. Forged as the Garden, which blooms with colour.
- **The Tide Staff (Atlantis).** A bronze staff with an Atlantean crystal held in fins of gold; water visibly swirls around it. Forged as the Sovereign's Staff with a crowned head.
- **The Abyssal Lure (Void).** A lantern made from an anglerfish lure and station brass, with a cold living light inside. Forged as the Deep Call, its light shifting and pulsing like a heartbeat.

Forge finish and skins

- The Pressure Forge finish is sea-glass: translucent frosted green-blue glass panels and inlays replacing parts of the metal, with light glowing through. Each Forged gun gets its new name engraved and a slightly altered shape (crushed, compressed, stronger-looking), not only a material swap.
- Alternate ammunition shows on the gun: Galvanic (copper coils on the barrel, sparks), Chum Rounds (red cartridges), Flare Burst (orange-tipped darts), Riptide (fins and vents at the muzzle), Boiling (a glowing heat jacket).

- Token skins (barnacle, verdigris, bone inlay, pearl, red tide) are material sets on every gun, with real geometry where the name implies it: barnacles sit on the surface, bone inlay is set into the stock.

Viewmodels, animation, and underwater effects

- Both hands always grip the gun correctly. A separate viewmodel FOV (55–65°). Aiming down sights aligns real sights or scopes with the eye.
- Every gun has draw, holster, idle sway (slower and floatier than on land), aim, fire, dry fire, partial and empty reloads, a Speed Brine reload 30% faster, and an inspect that shows the Forge finish, exactly as the design doc's Animations section lists.
- Reloads are mechanically true: gas pistols pop the cylinder and thumb in darts one at a time; needlers swap a glass cartridge with a hiss of bubbles; powder guns eject a sealed brass clip that sinks and glints; spearguns cock with both arms straining; harpoon rifles winch a bolt from the quiver; the scatter gun loads shell by shell.
- Firing underwater: gas guns release a burst of bubbles from vents; powder guns give a brass flash and a shock ring in the water; spears and bolts leave a cavitation trail and a line snapping taut. Recoil is damped and drifts as in water. Spent cases and clips sink with physics.
- Hits on beasts release blood ribbons and scales; misses kick up silt or chip coral.
- The third-person model of every gun (held by teammates and enemies) shows the same finish, attachments, and Forge state at lower detail.

Fish, beasts, and flora

All ~200 beasts and ~60 flora must read as their real species at a glance, look alive in motion, and show the ecosystem's state (hungry, fed, wounded, fleeing, dead) on their bodies. Use the creature kit from section 3 for the common work and give hero species individual attention.

Anatomy standard

- Correct body plan and proportions per species: fin count and placement, gill slits or plates, the lateral line, eyes set where the species has them, mouths that open to the right shape, teeth, barbels, spines, and claws.
- Eyes are separate meshes with a wet specular highlight and, for predators, a reflective tapetum that glints in diver lamps.

- **Surfaces:** scales as a normal-map pattern sized to the species, iridescence on silver fish, slime sheen on eels and amphibians, shark skin as fine denticles, shells with growth rings and encrusting life, cephalopod skin with chromatophores that animate colour and pattern.
- Each species record's 3-band colour (base, belly, accent) from the design doc drives the material, plus a species pattern (stripes, spots, bars, mottling) authored once per species.
- Size class sets detail: size 1–2 are clean and readable; size 3–4 add scars, parasites, and individual markings; size 5–6 get hero-level sculpting.
- Weak points from the design doc's damage model are modelled visibly: gills that flare open, soft pale undersides on shelled beasts, the cephalopod vent, reptile throats.

Hero species to model individually

At minimum, give individual sculpting and animation to every apex (tier 4) and every unique map resident: the sharks (reef, hammerhead, great white, Sleeper Shark), barracuda, morays, the grouper family, the salon's crocodile, the octopus, the orca pod and the Matriarch's pod, mantas, whales, dolphins and seals, sea turtles, the bobbit worm, the giant squid, the water monitor, temple kraits, the Pale Caiman, coelacanths, yeti crabs, the Ghost Worm, the Sand Worm, the Relict, and the Alien Horror. The rest come from the kit with species tuning.

Animation by body plan

Keep the design doc's animation rules (sine along the spine, flapping rays, undulating eels, IK crab legs, Chains for cephalopod arms, four-limb gaits), but author them properly in Blender as actions and blend them in-engine:

- **Fish.** Idle hover with fin flutter, cruise, burst swim, sharp turn, feed (strike and swallow), flee, a hurt flinch, and death (rolling belly-up and sinking or floating by species).
- **Sharks.** A slow, heavy cruise; the pectoral-fin drop and arched back as a threat display; a jaw-protrusion bite with the eyes rolling back.
- **Eels and morays.** Undulation, a strike from a hole and a pull back, the jaws working constantly.
- **Rays and mantas.** Wing flaps, gliding, the burrow-and-burst from sand.
- **Crustaceans.** IK walking, claw threats and snaps, defensive retreat into a hole, shells that crack.

- **Cephalopods.** Jetting, arms trailing, crawling, ink release, colour flashes and camouflage that blends into the surface beneath.
- **Reptiles, amphibians, mammals.** Four-limb swimming, surfacing to breathe in air pockets, death rolls (crocodilians), the whale's slow tail beat.
- **Schools (boids).** Instanced, each fish with a small phase offset; the scatter and regroup from the design doc is the school bursting outward in silver flashes.

States shown on the body

State	What the player sees
Hungry or hunting	Faster tail beat, mouth slightly open, colour darkening (cephalopods flash), eyes tracking the diver
Fed	Distended belly, slow cruise, settling to rest
Wounded	Visible wounds where hit, blood ribbons streaming from them (following the scent grid's flow as the design doc says), torn fins
Fleeing at low HP	A blood trail behind it, erratic swimming, fins clamped
Netted	The net mesh wrapped around the body, struggling
Parasite attached	The parasite model clinging and pulsing
Camouflaged	Pattern matched to the surface beneath; a faint shimmer within 2 m
Dead	Eyes clouding, colour draining over 10 s, sinking or floating; scavengers tearing pieces off the corpse over time

Flora

Flora is part of the ecosystem and often a hazard, so it gets creature-level care: kelp with flexible bones that sway in current and part around divers, sea grass that bends, anemones that retract when touched, grabbing coral (the Reef's Reacher gates) with

visible muscle, bacterial mats with a sickly sheen, glow-moss that pulses, glass sponges with fine lattice detail, and black coral. Grazed flora visibly thins; trampled flora bends and recovers.

Enemies and bosses

Each faction is a fully designed group of characters with its own materials, silhouettes per unit type, and a look that tells the player what it is and how it fits the food web. Each boss is the hero creature of its map and gets the most modelling time of anything in the game.

Faction design

Faction units use the shared humanoid rig (or their own rigs where they are not human). Every unit type in a squad must be identifiable by silhouette at 30 m, because tactics depend on spotting the Netman, the Lantern Bearer, or the Shaman first.

Faction (map)	Look and materials	Unit silhouettes	Food-web tells
The Wreckers (Sunken Ship)	Salvage pirates in patched, mismatched diving suits; rusty helmets with welded grilles; tattoos and gold teeth visible through ports; chum buckets on belts	Cutter: lean, two knives, fast crouched swim. Speargunner: long speargun. Netman: net folded on the back, weighted lines. Foreman: big, armoured, a Harpoon Cannon on the shoulder	They bleed like beasts: wounds stream blood; they panic visibly when sharks arrive
The Drowned (Cave)	Ghost sailors: pale translucent bodies inside rotted period clothing, drifting hair, empty glowing eyes, lanterns on poles	Deckhand: boarding axe, hunched. Lantern Bearer: a tall pole with a bright lantern. The Bosun: a cutlass ghost who phases through stalagmites with a distortion effect	No blood. Their lanterns throw real light and visibly draw curious species

Faction (map)	Look and materials	Unit silhouettes	Food-web tells
The Reef Raiders (Reef)	An invented island tribe of free-divers: carved bone and shell ornaments, woven gear, patterned body paint, riding sea turtles	Spear Diver: long spear. Turtle Rider: on a saddled turtle. Sling Diver: a sling and coconut bombs. Shaman: a drum, a feathered and shell headdress	They carry meat and trail blood from their kills. The Shaman's drum sends visible pulses through the water
The Lost Ones (Atlantis)	Drowned Atlantean citizens who breathe water: grey-green skin, gill slits, bronze and gold armour, tattered robes	Legionnaire: gladius and tower shield, phalanx-ready. Cultist: robes, a void-glowing amulet. Slinger: light armour. Armored Lost One: heavy plate. Priest: a tall crown and staff	Black ichor instead of blood: a distinct dark, oily ribbon
The Remnant (Void)	A drowned research station's survivors: pressure-suited researchers, brass clockwork sentinels, void cultists	Sentinel: clockwork, rotary needlers, exposed gears. Researcher: a bulky pressure suit, gas grenades, a medical pack. Cultist: robes over a diving suit. The Station Chief: a size-5 diving mech	Sentinels leak oil (iridescent slick, no scent); the others bleed normally

Shared enemy needs. Squad formations, flanking, suppressing fire, taking cover, and retreating as the design doc describes, with readable body language: pointing, signalling, reloading behind cover, panicking when a beast attacks. Each faction gets bark animations to match its barks. Enemies dropped weapons use the same gun models as the players' guns. Faction entries (the hull breach, the cave mouth, the drop-off, the city gate, the airlocks) get visual set-dressing so players learn to watch them.

Bosses

Every boss gets a hero sculpt, a full custom rig, and animations for every attack and phase in the design doc. The expose and the weak point must be visually obvious once the player knows to look.

Boss (map)	Visual design	What must read
The Goliath (Ship)	A colossal grouper-like fish with heavy armoured plates on the head and flanks, barnacles, old harpoons and chains embedded in it	The gills behind the plates flare open and glow red during the 2 s windows; the Inhale shows water rushing into its mouth; it stops being lit when the algae mat is shot off
The Lobster (Cave)	A giant cave lobster, pale and eyeless from life in the dark, with crystal coral grown on its shell	Rearing up to show a soft, veined underside when the crystal coral rings; stalactite impacts that crack the shell
The Matriarch (Reef)	An enormous orca matriarch, scarred and old, leading her pod	Her charges in straight lines; flipping and exposing her belly when the surge channel hits her; pod members marked so the team can find them
The Cistern Wyrms (Atlantis)	A huge sea serpent in Atlantean colours, with bronze fittings once used to tame it	Stranded and writhing when the god-pool drains; surfacing at the fourth grate; its blood filling the aqueduct
The Lantern Leviathan (Void)	A giant deep-sea anglerfish-like apex with a vast mouth, dark skin, and a pale lure	The lure as the bright weak point; the lure splitting into three in phase 3; the reactor scald showing as burns

Each phase transition changes the boss visibly: plates crack, wounds open, colour shifts, behaviour gets frantic. Phase 3 enragement shows in posture, eye glow, and speed. Bosses that reform after 6 minutes come back scarred.

The maps

Each map becomes a dense, handcrafted-looking place built from a Blender modular kit and placed by the existing layout data. The zones, routes, doors, and costs in the design doc stay exactly where they are. Every zone gets a landmark visible from its entrance, so players learn the map by sight.

Rules for every map

- **Big shapes, then small detail.** Strong readable architecture or rock forms first, then dense layered set dressing: weed, barnacles, sediment, debris, flora, and drifting particles.
- **Interactables stand out.** Racks (weapons chalked on walls), Davy's Locker (with its lantern buoy), tonic machines, the Forge, workbenches, doors with their scrip prices, traps, and power switches each get a distinct, well-lit, detailed model that never gets lost in the decor.
- **Habitats look like habitats.** Where a species lives (the crocodile's lair, the moray pipes, the eel holes, the cleaning station) the set dressing shows it: bones, nests, cleared sand, shed shells.
- **The map responds.** Blood drifts with the current; shot coral chips; corpses sink and get picked over; doors bought open stay open with debris pushed aside.

Per-map treatment

The Sunken Ship. The Marguerite as a real 60 m steam yacht on her side at 30 m: hull plates with rivets, a funnel, masts and rigging trailing kelp, portholes throwing light shafts. Inside, the tilt matters: furniture piled against the low walls, the grand salon's chandelier hanging sideways, the air pocket under the tilted ceiling with a silver surface. Rotting mahogany panelling, peeling wallpaper, tarnished brass, a velvet-and-gilt salon, a tiled galley, cabins with personal belongings, a riveted engine room with the generator and the Goliath's nest. Outside: the sand shelf, the keel, and the sharks' approach lane fading into blue. The foredeck crane and container trap are visible from the salon.

The Underwater Cave. Wet black and grey limestone with layered strata, flowstone, columns, stalactites and stalagmites, silt floors that cloud when disturbed. The Mouth's single light shaft; the Flooded Gallery's columns; the Chimney's vertical shaft with ledges; the Dry Chambers in air with dripping water, ripples, insects, and amphibians; the Sump's

silt and bacterial mats; the Dynamo Sump's drowned expedition generator with cables; the Cathedral's vast space with glowing crystal coral. Slipstreams are visible as rushing particle currents and bubble streams with a clear entry mouth.

The Coral Reef. The showcase map: dense, layered reef built from many coral species (staghorn, brain, table, fan, soft corals), with fish everywhere. The Lagoon's mangrove roots and bright shallows, the Staghorn Forest's corridors, the Bommie with its cleaning station, the open Sand Flats with sea grass and garden eels, the Brain Coral Maze, the Blue Hole's dark circular drop, and the Drop-off Wall falling into deep blue where mantas, orcas, and whales pass. Sunlight caustics everywhere; colour falls off to blue with depth.

Atlantis. A real city plan on a 120 m hill, 500 m across, with 60+ buildings: the great wall and gatehouse, terraced farms with irrigation canals and a mill, the lower town's houses, market, and toolsmiths' street, bathhouses with hypocausts, barracks and the armory, the library with its observatory, the forum and government hall, and the grand chapel with the god-pool on the summit. Marble and sandstone with mosaics, bronze statues, and columns, all overgrown: algae, sponges, sea fans, coral growing out of windows, silt drifts in streets. The aqueduct as a landmark running down the hill. The lighthouse on the wall. The crystal engine glowing in the treasury.

Approaching the Void. The Rim: a vast black sand plain with sea pens and sand eels, the worm's tunnels on the horizon, and the abyss as an enormous black edge. Below: vertical chimneys and galleries with glass sponge reefs, bamboo coral, black coral, tube worms, and hydrothermal vents in the Warrens. The research station: a rusted early-20th-century structure of riveted steel modules, portholes, specimen tanks (some broken), terminals, the reactor, and the mech bay. The Lowest Vault: a cathedral cave open to the abyss. The void itself: a black column with drifting motes and a faint pale light far below, as the design doc says, which must look vast and wrong.

Traps and dynamic set pieces

Every trap in the design doc gets a full animated set piece: the cargo container dropping across the salon door, the Chimney's stalactites falling, the Reacher coral gate closing and eating, the surge channel flushing the Bommie, the canal sluice flooding a terrace, the hypocaust boiling a room (steam and bubbles), the god-pool draining and refilling, the pressure door crushing a corridor, the reactor vent's scalding jet, and the vault ledge collapsing into the abyss.

Rendering, effects, and UI

The renderer must make water the main material of the game: light that scatters and fades with distance and depth, caustics near the surface, and particles everywhere. It builds on the shared PBR renderer from The Trawl spec, extended in raylib with custom shaders.

Underwater light.

- Distance and depth fog with per-channel absorption (red disappears first, then green), so colour shifts toward blue-green with distance and depth, tuned per map and zone.
- Volumetric light shafts from the surface, portholes, and openings, plus caustics projected onto surfaces near the surface (the design doc's caustics stay; make them animated and soft).
- PBR shading with cascaded shadows from the sun shaft where it exists, and point and spot lights with shadows for diver helmet lamps, station lights, lanterns, muzzle flashes, and bioluminescence.
- Screen-space ambient occlusion and baked AO so creatures and divers sit in the scene.
- Bioluminescence as emissive materials with bloom, which matters most in the Cave and the Void.

Particles and water effects.

- Marine snow everywhere (the design doc's rule), denser in the deep.
- Bubbles from helmets, guns, slipstreams, vents, and struggling creatures, rising and wobbling.
- Silt clouds when divers or beasts touch the bottom; shells and coral chips from hits.
- Blood as ribbon particles following the scent grid's flow (from the design doc), dark red and thick in clear water, and black ichor for the Lost Ones, iridescent oil for Sentinels.
- Air pockets with a mirrored, rippling surface from below, and a dry, dripping world above.

Post-processing. HDR, filmic tonemapping, bloom on bright sources, a gentle lens distortion and vignette at the helmet edges, chromatic fringing at the very edge of the port, and the design doc's red tint at the mask edge as local scent rises. The refined ink line from section 2 is applied here, after fog, so it fades with the water.

HUD. Keep the design doc's brass-helmet HUD and make it a real model: the inside of the helmet rendered with brass, rivets, and glass, gauges as real dials with needles (the scent meter, the predator pulse, air, scrip as a mechanical counter), condensation and droplets on the glass, and cracks when the diver is badly hurt. Menus, the dossier, and the how-to-play page get matching period design: aged paper, brass, typewriter and engraved type.

Accessibility and readability. A brightness and fog-density calibration screen; a colourblind setting for blood, ichor, and scent colours; and a motion-comfort setting for viewmodel sway and lens effects.

Process and acceptance

Work in ten phases, building one complete map (the Sunken Ship) to full quality before scaling to the other four. Each phase ends with before/after renders that Ray approves before the next starts.

Screenshot harness. Extend the design doc's `--reddtide-shots` (every creature, weapon, and map zone to PNG) to render at 1920×1080 with the new renderer, from fixed cameras. Also render turntables from Blender for every new asset before it enters the game. Claude Code looks at both, compares them with this spec, and fixes problems before showing Ray. The fixed shots are:

- 1. Every map zone from its entrance, showing its landmark.
- 2. Each diver: front, side, back, a face close-up through the helmet port, and each skin.
- 3. Each gun: side and three-quarter views on a neutral background, the viewmodel at rest, aiming down sights, and mid-reload, plus the Forged version.
- 4. Each faction: a full squad lined up, then the same squad in combat.
- 5. Each boss: a full-body hero shot and one shot per phase.
- 6. Each hero species, plus a sample of kit species, in its habitat.
- 7. A busy combat moment per map: four divers, a squad, beasts, blood in the water.

Phases.

Phase	Work	Done when
1. Pipeline	Blender connection, build scripts, glTF export, raylib loading with skinning and instancing, the material library, the harness	A test gun, a test diver, and a test fish load in-game with full materials and animation

Phase	Work	Done when
2. Rendering	Underwater light, fog, caustics, shafts, particles, post, the refined ink line	One Sunken Ship room looks finished with placeholder props
3. Divers	The four suits, bodies, faces, skins, animations, first-person arms, the helmet HUD	Shot set 2 passes the diver criteria
4. Guns	Rack guns, melee, Locker guns, the Forge finish, skins, all animations and effects	Shot set 3 passes the gun criteria
5. Creature kit	All 15 body-plan rigs, the species-record script, the states, flora	20 sample species pass the creature criteria
6. Sunken Ship complete	Ship kit, all zones, the Ship's species, the Wreckers, the Goliath, the Galvanic Rod	The whole map passes every criterion: this is the vertical slice
7. The Underwater Cave	Map, species, the Drowned, the Lobster, the Resonator	Map passes
8. The Coral Reef	Map, species, the Reef Raiders, the Matriarch, the Anemone Gun	Map passes
9. Atlantis	Map, species, the Lost Ones, the Cistern Wyrms, the Tide Staff	Map passes
10. Approaching the Void	Map, species, the Remnant, the Lantern Leviathan, the Abyssal Lure; final polish and performance	Map passes; the whole game meets the performance target

Diver criteria.

- Each diver is identifiable from silhouette alone at 40 m, in every skin.
- A face is visible and expressive through the helmet port.
- Gear reads as real brass, copper, canvas, rubber, and steel with salt wear.
- First-person hands grip every item correctly.

Gun criteria.

- Every gun has the parts in its per-weapon row, with every moving part animated.

- Each gun looks like a plausible machine for firing underwater (seals, gas, valves, cartridges visible).
- The gun is the most detailed object in its frame.
- Sights align when aiming down sights.

Creature criteria.

- A player who knows the real animal can name its species at a glance.
- Weak points are visible.
- Hungry, wounded, fleeing, and dead states read without the HUD.
- Schools look alive (no lockstep motion).

Map criteria.

- No surface is a bare untextured primitive.
- Every zone has a landmark recognisable from its entrance.
- Racks, the Locker, tonics, the Forge, doors, and traps stand out from set dressing.
- Each map matches its mood row in section 2.

Performance (decided). The target machine is the computer the game is being coded on. In phase 1, Claude Code reads that machine's hardware (CPU, GPU, VRAM, RAM, screen resolution), records it at the top of the build notes, and holds the design doc's target on it: 60 fps at 1080p with four players and 300 active agents. Use instancing, LODs, billboards past 80 m, texture atlases, and occlusion culling per zone. If a feature costs too much on that machine, report it with frame-time numbers and propose a cheaper version.

Decisions from Ray.

- **Ink line:** keep it where possible; Blender-built detail takes priority (section 2).
- **Reef Raiders:** acceptable as written; no redesign.
- **Performance target:** the computer the game is coded on (above).

Kickoff prompt for Claude Code

Paste this into Claude Code with the link to this doc and current Red Tide screenshots attached.

```
Read "Red Tide — Visual Overhaul Spec" in full, then the Red Tide design doc's Presentation and technology section. The spec replaces the design doc's Art section and the no-assets rule
```

for Red Tide visuals; every system, number, and rule in the design doc still stands. It changes how the game looks, never how it plays.

Use the connected Blender tool for all modelling, texturing, rigging, and animation, and follow the spec's Blender pipeline exactly: every approved asset also gets a committed Python rebuild script and .blend file, and is exported to glTF for raylib.

The overhaul covers everything: the four divers, every weapon (the most realistic objects in the game), all beasts and flora, the five factions and five bosses, all five maps, and the underwater renderer. Build the renderer, rig, viewmodel system, gun-part kit, fish rigs, and screenshot harness as shared engine code with The Trawl.

Decisions already made (see the spec's Decisions from Ray): keep the refined ink line, but Blender-built detail takes priority over it; the Reef Raiders stay as written; the performance target is this computer, so read and record its hardware first.

Start with phase 1 only: confirm the Blender connection works by building and rendering one test asset, then build the pipeline and harness. At the end of each phase, show me before/after renders, check them against the acceptance criteria yourself, and wait for my approval before the next phase. The Sunken Ship is the vertical slice: get it fully finished before starting the other maps. If anything in the spec is impossible in raylib or too expensive on this computer, say so with numbers and propose an alternative instead of quietly doing less.